Answers Microeconomics

Part A

1. (a) Alesha's utility is concave in x. The first-order condition is $8 - x = 0$ so she chooses $x^* = 8$. Her utility is $u_A(8, 32) = 64 - \frac{64}{2} + 32 = 64$. Bob's utility is $u_B(8, 32) = -\frac{64}{2} + 32 = 0$.

   (b) As the utility functions are quasi-linear we can add them and choose x to maximize $8x - x^2$. This is concave, and the first-order condition is $8 - 2x = 0$ so $x^{PE} = 4$. Equivalently the marginal benefit of x is $8 - x$ and the marginal damage caused by x equals x and we can equate marginal benefit and marginal cost. More formally, and probably only the best students will do this, we choose x and Alesha's income (with the remaining income going to Bob) to maximize $u_A(x, y_A)$ subject to the constraint that $u_B(x, y_B)$ equals a fixed level $\bar{u}$. The Lagrangean is

   $$8x - \frac{x^2}{2} + y_A + \lambda \left[-\frac{x^2}{2} + 64 - y_B - \bar{u} \right].$$

   The value of the Lagrange multiplier is 1, which is what justifies adding the utility functions.

   (c) Denote the tax by t. Alesha's first-order condition becomes

   $$8 - x - t = 0.$$

   Set $x = 4$ in this first-order condition, which implies that the tax must be 4. Tax revenue is $16 = 4 \times 4$. The tax equals marginal damage at the Pareto efficient level of x.

   (d) Remember that Alesha pays the tax, so her income goes down by 16. Her utility is

   $$8 \times 4 - \frac{16}{2} - 16 + 32 + C = 40 + C$$

   where C is the compensation. Setting this equal to her utility before the tax of 64 yields $C = 24$. So giving her back the tax that she pays of 16 isn't enough. Returning just the tax revenue compensates her for the tax payment, but she still loses because x is below her unconstrained optimal level. Thus Bob needs to pay a lump-sum tax of 8, and this and the tax revenue need to be given to Alesha to ensure that she is fully compensated.

2. (a) If a also sets $p_a = 6$ then it has profit of zero (as in a pure Bertrand model). Here, however, an increase in price will not drive a's demand to zero, and will now yield positive profit. Suppose $p_a = 6 + \varepsilon$. a's profit is $\varepsilon \times (6 - 6 - \varepsilon + \frac{6}{2}) = \varepsilon(3 - \varepsilon) > 0$ for $\varepsilon \in (0, 3)$. (This is optimized with $\varepsilon = 1.5$ so $p_a = 7.5$, though this is not required.)

   (b) Firm a chooses p_a to maximize $(p_a - 6)(6 - p_a + 0.5p_b)$ (which is concave in p_a). The first-order condition is

$$6 - p_a + 0.5p_b - p_a + 6 = 0$$

   so the best response is

$$p_a = \frac{12 + 0.5p_b}{2} = 6 + \frac{p_b}{4}.$$

   b's best response is

$$p_b = 6 + \frac{p_a}{4}.$$

   Solving simultaneously (or noting that it is clear that the equilibrium is symmetric) gives

$$p_a = p_b = 8$$

   as required. As candidates are given the answer it would also be acceptable to show that each firm setting 8 satisfies the best responses.

   The profit margin is 2. Output per firm is $6 - 8 + 0.5 \times 8 = 2$. So profit per firm is 4 and total profit is 8.

   (c) The merged firm will certainly take advantage of the ability to pay 20 to reduce marginal cost to zero. If the merged firm kept each price equal to 8 then it would continue to sell 2 units of each good. Without the cost reduction total cost would be $2 \times 2 \times 6 = 24$ (two firms each selling two units at a marginal cost of 6). With the cost reduction total cost is just the fixed cost 20.

   The merged firm is a monopolist that chooses p to maximize $(p - 0)2(6 - 0.5p) = p(12 - p)$. The first-order condition is

$$12 - 2p = 0,$$

and profit is concave in the price, so the monopoly price is 6. Monopoly profit, while paying the lump sum of 20, is $(6-0)(12-6)-20=16$. The merged firm makes more profit, 16, than the two independent firms did, 8. Consumers obtain a lower price and so are better off. Thus a competition authority would allow this, whether it cares about social welfare (consumer surplus plus profits) or just about consumer surplus. Note that the large reduction in marginal cost from 6 to 0 induces only a small reduction in price from 8 to 6, because of the effect of increased market power.

Some might mention Upward Pricing Pressure. (Not required, though it can be applied to show that the firms want to cut the price.) Firm 1 will want to increase its price in a merger with 2 if

$$\text{2's pre-merger margin} \times \text{Diversion Ratio}_{12} > \text{Fall in } MC$$

This is not satisfied here. The initial margin is 2 and the diversion ratio is 0.5. Their product is $2 \times 0.5 = 1$, which is below the marginal cost reduction of 6.

3. (a) Under observable effort A participates and works hard if the wage contingent on effort is at least 14 because in this case his net income is at least $14-5=9$, better than his outside option. P's expected profit from offering $w=14$ for $e=1$ (and no wage for $e=0$) yields an expected profit of $\frac{1}{2}60+\frac{1}{2}20-14=26$. P prefers this rather than not hiring A and getting 20 for sure.

(b) If A accepts this revenue-sharing contract then he chooses $e=1$, because then he gets an expected utility of $\frac{1}{2}\sqrt{30-5}+\frac{1}{2}\sqrt{6-5}=3$, which is strictly greater than his expected utility from $e=0$ under the contract, $\sqrt{6}\approx 2.45$. He accepts the contract in the first place because the utility of his outside option is slightly less than $\sqrt{9}=3$. The net expected profit for P will be $\frac{1}{2}(60-30)+\frac{1}{2}(20-6)=22$, still better than not hiring A.

(c) In the optimal effort-inducing revenue-sharing wage contract $\{w_{60}, w_{20}\}$ the participation and incentive constraints both bind. If the IR constraint is slack then both w_{60} and w_{20} can be reduced without upsetting incentives; if IC is slack like in in part (b) the agent is exposed to too much risk.

The equations to solve are $\frac{1}{2}\sqrt{w_{60}-5}+\frac{1}{2}\sqrt{w_{20}-5} =_{\text{IC}} \sqrt{w_{20}} =_{\text{IR}} \sqrt{9}$.

From this $w_{20}=9$ and $w_{60}=21$.

Hence P's expected profit is $\frac{1}{2}(60-21)+\frac{1}{2}(20-9)=25$, which is still better than not hiring A.

(d) The agency cost is the difference between P's first-best profit under observable effort in part (a), and the second-best optimal revenue-sharing contract in part (c), that is, $26-25=1$ monetary unit.

4. (a) By buying x amount of coverage at unit price $\frac{1}{2}$ Cornelia reduces her final wealth from 10 to $10 - 0.5x$ in case of no loss and increases it from 2 to $2 + 0.5x$ in case of loss. She maximizes

$$EU(x) = \frac{2}{3}\ln(10 - 0.5x) + \frac{1}{3}\ln(2 + 0.5x)$$

and the (by concavity sufficient) FOC of her problem is

$$\frac{2}{10 - 0.5x} = \frac{1}{2 + 0.5x} \Leftrightarrow 4 + x = 10 - 0.5x \Leftrightarrow x = 4.$$

She buys 4 units of coverage; her final wealth will be 8 (no loss) or 4 (loss).

(b) Competitive insurers would set $\bar{\pi} = \frac{1}{3}$ (fair premium, or zero expected profit) and Cornelia would choose full coverage, $x = 8$. Candidates should explain why this is the case, e.g., saying that at fair premium full coverage second-order stochastically dominates Cornelia's stochastic final wealth at any other coverage level.

(c) The competitive screening contract in insurance is such that Cornelia with $\frac{1}{2}$ loss probability is indifferent between full coverage at premium $\pi = \frac{1}{2}$ and limited coverage $\bar{x}$ at premium $\bar{\pi} = \frac{1}{3}$. (In this case Cornelia with $p = \frac{1}{3}$ strictly prefers the latter to the former.)
Full insurance at $\pi = \frac{1}{2}$ yields a sure utility of $\ln(6)$. Therefore $\bar{x}$ must satisfy

$$\frac{1}{2}\ln(10 - \bar{\pi}\bar{x}) + \frac{1}{2}\ln(2 + (1 - \bar{\pi})\bar{x}) = \frac{1}{2}\ln(10 - \tfrac{1}{3}\bar{x}) + \frac{1}{2}\ln(2 + \tfrac{2}{3}\bar{x}) = \ln(6).$$

By substitution $\bar{x} = 3$ works, $\frac{1}{2}\ln 9 + \frac{1}{2}\ln 4 = \frac{1}{2}\ln 36 = \ln 6$, whereas $\bar{x} = 2, 4$ do not.

(d) Cornelia is better off in the screening contract with coverage $\bar{x} = 3$ at premium $\bar{\pi} = \frac{1}{3}$ because then her final wealth is 9 (no loss) or 4 (loss), as compared to 8 (no loss) or 4 (loss) in the solution in part (a). Insurance companies are worse off because they no longer make a profit on her insurance.

Part B

5. **EITHER**

Candidates should first explain the two theorems: the First says that Walrasian equilibria are Pareto efficient, the Second that any Pareto-efficient allocation can be achieved with a competitive economy by reallocation of initial endowments or a system of lump-sum taxes and transfers.

There should be a brief discussion of the conditions under which each theorem holds: the First requires no externalities, or more generally no missing markets, and the Second requires convex preferences. Proofs of the theorems are not required or expected.

Much economic policy is predicated on the first theorem, and with a hidden assumption that appropriate redistribution of income is going on in the background, and treats deviations from the necessary conditions as correctable. For example policy towards externalities suggests imposing a Pigouvian tax equal to marginal damage at the optimum, but not taking account of any effects on income distribution. Competition policy aims to ensure a reasonable level of competition, clearly a necessary condition for the first theorem to apply.

The second theorem suggests that correction of market failure can be separated from income distribution. Of course real-world taxes are not lump-sum and generate deadweight losses, so it is not clear that such a separation applies in practice. Candidates are not expected to know about models of optimal income taxation (though some will know such models).

OR Three models of trade are covered in the lectures.

(1) The Ricardian model of comparative advantage has a single factor, labour, with constant productivities in each of the industries. There will be complete specialization and all agents will be better off: real wages will increase. Independent of tastes.

(2) Within the Heckscher-Ohlin framework (only touched on) there is the Stolper-Samuelson theorem (which is fully covered). Free trade, which leads to an increase in the price of the exportable good, causes a rise in the real return to the factor used intensively in the exportable industry and a fall in the real return to the factor used intensively in the importable industry. Assuming that factor ownership is unequal (so, for example, there are households who rely on labour income and have little if any capital ownership) the introduction of free trade will harm the factor used intensively in the importable industry

and benefit the other. Because factors are fully mobile across industries it does not matter which industry a factor, say labour, actually is employed in.

(3) The general model of trade, with a stricly convex production set and thus incomplete specialization, suggests that output of the exportable rises and that of the importable falls. If income distribution is equal then all are better off, but if factor ownership is concentrated, as implicitly in Stolper-Samuelson, then some could lose. Consumption effects of the move to free trade: suppose a single consumer, this person is better off as their consumption possibilities exceed their production possibilities. Consumption of at least one good will increase.

6. A (pure) public good is non-rival and non-excludable. Four topics that might be covered. (i) Private provision of a public good: if marginal valuations differ then only the agent with the highest marginal valuation provides the public good in the Nash equilibrium with private provision. The remaining agents free-ride. (ii) The Samuelson condition: the Pareto-efficient level of provision is where the sum of the marginal benefits equals the marginal cost. (iii) The Lindahl equilibrium has each agent paying a different price, with the sum of the prices equalling marginal cost. This ensures that Samuelson condition is satisfied, and the public good is financed. But it presumes that the government knows the marginal values. (iv) The Clarke-Groves scheme: provided appropriate transfers can be made, agents can be induced to tell the truth. The pivotal agent scheme ensures that there is no need for a net contribution from the government. But participation is not guaranteed, and a fundamental issue is that agents might not know their own valuations.

7. How can the theory of repeated games be used to help to identify the factors that determine whether collusion in an industry is sustainable or not? Is it a problem that repeated games may have multiple equilibria?

   A simple model has n price-setting firms, identical products, no capacity constraints, constant and equal marginal costs and infinite repetition. Strategies that reward cooperation and punish cheating will form equilibria provided the discount factor (the weight on profits one period ahead) is large enough. The grim trigger strategy works: price at the monopoly price as long as everyone has done so thus far, set price equal to marginal cost for ever if anyone cheats. These form a sub-game perfect equilibrium if the discount factor, $\delta < 1$, satisfies
$$\delta \geq \frac{n-1}{n}.$$
   If n increases then the right hand side rises and the inequality is less likely to hold. If δ rises, e.g. because the real interest rate, r, falls, with $\delta = \frac{1}{1+r}$, then the condition is more likely to hold. If the frequency of sale falls, say there is a sale every other period, then the condition above becomes $\delta^2 \geq (n-1)/n$ which is less likely to hold because $\delta^2 < \delta$. Ease of detection: if cheating can go on for two periods before being detected then the same condition applies as with frequency of sales.

   Multimarket contact: if firms meet in one market with sale every period, and another with sales every other period, then cheating in the latter market can be more rapidly met by punishment in the first market, making collusion easier to sustain.

Several factors (cost asymmetries, existence of meeting competition clauses, price transparency, existence of leniency programmes) affect collusion but are not particularly related to the repeated games framework.

Game theoretic problem: it is true that there are multiple equilibria with repetition: e.g. if all firms have the strategy to price at MC forever that is an equilibrium. Another strategy is to price halfway between the monopoly price and marginal cost, and to punish if someone cheats. This would be Pareto dominated by all pricing at the monopoly price (with the threat of retaliation remaining). The theory suggests that if the discount factor is large enough then there will be collusion forever. So no player ever observes punishment, and how will they know that the other players are playing the required strategy?

8. Start with the content of EU theory: under certain axioms on preferences over monetary lotteries (complete, transitive, continuous preferences satisfying the independence axiom) an individual behaves as if they maximized the mathematical expectation of a utility function $u(w)$ on final wealth $w \in \mathbb{R}$.

   The individual is risk averse if they prefer the expected value of a lottery received for sure over the lottery itself: $\mathrm{E}[u(W)] < u(\mathrm{E}[W])$, where W is a random variable representing a lottery over final wealth. Risk aversion is equivalent to concavity of u by Jensen's inequality.

   Risk aversion does not preclude investment in a risky asset as long as the asset has a positive expected return. This can be illustrated using the binary-outcome insurance framework, for example.

   Suppose that Ann with initial wealth w may take on any fraction $q \in [0, 1]$ of a project that gains £1 in state N and loses £1 in state A; let the probability of state A (loss) be $p < 0.5$. By choosing fraction q her expected utility is $pu(w - q) + (1 - p)u(w + q)$; her optimal q satisfies the FOC $-pu'(w - q) + (1 - p)u'(w + q) = 0$. At $q = 0$ the left-hand side of the FOC is positive, $-pu'(w) + (1 - p)u'(w) = (1 - 2p)u'(w) > 0$, because the expected return on the project is positive, $p < 0.5$. Hence Ann invests $q > 0$. A somewhat less formal argument would be to say that there is no exposure to risk at $q = 0$, hence Ann behaves as if risk neutral and invests at least a little bit as long as $p < 0.5$.

More risk-averse agents invest less. With DARA preferences a richer agent invests more. Well-read candidates may discuss the effects of stochastic initial wealth (under DARA it results in less investment, precautionary savings effect), portfolio diversification, other issues.

9. A more regulated, apparently "safer" road reduces motorists' incentives to drive carefully (more slowly and/or while paying more attention). The net result may be more accidents. This is a classical "insurance v incentives" trade-off. Fundamentally it is an empirical question whether more regulation makes road users safer after their behaviour adjusts. A good answer may sketch a model with hidden costly effort and road regulation as substitutes.

   The observation that perception of safety increases risk taking ("good brakes make a car go faster") is called the Peltzman effect. Sam Peltzman (U of Chicago) in his 1975 paper "The Effects of Automobile Safety Regulation" argued that highway safety regulations did not reduce highway deaths. He found that mandatory safety belts increased the number of accidents but reduced their average severity with the overall death rate staying constant. Many other empirical studies followed with varying results.

   The question does not ask candidates to think in terms of adverse selection or signalling.

   Some might say that excessive safety regulation (or self-driving cars, etc.) reduce road users' private investment into becoming better drivers. These are interesting dynamic, investment effects. Some might also mention the intrinsic value of allowing people make socially responsible decisions in a free society. The question explicitly asks about moral hazard so these are less good (but otherwise interesting) answers. There was an article by Simon Jenkins ("The removal of road markings is to be celebrated. We are safer without them") in 2016 making related points.

   Note that traffic calming/ slowing measures will also impose a cost on those who want to get to their destination faster. That is a cost that most traffic engineering studies do not consider at all.

10. (a) Conditions (i)–(ii) are equivalent to $t = 0.5x + 0.5y$ and $w = 0.8x + 0.2y$. Two linear restrictions on four parameter values imply two parameters are still free. EU payoffs have to be invariant to any positive affine (constant+linear) transformation, which is an additional restriction, so only one parameter is free. If $t = 5$ then we can still set either $x > 5 > y$ or $x < 5 < y$, implying different preferences for player 1 over final outcomes. Therefore we can normalize one parameter only (e.g., $t = 5$).

(b) R is strictly dominated for player 2, by e.g., $\frac{1}{2}L + \frac{1}{2}C$. Player 1 has no dominated strategy even after R is eliminated: if $x < y$ then B is best reply to C and T to L; if $x > y$ then T is best reply to C, B is to L.

Payoff tables with specific parameter values as set in parts (c)–(e):

$x = 0$	L	C	R
T	$\underline{5}, \underline{5}$	$0, 1$	$\underline{10}, 2$
B	$2, 2$	$\underline{10}, \underline{6}$	$0, 3$

$x = 10$	L	C	R
T	$5, \underline{5}$	$\underline{10}, 1$	$0, 2$
B	$\underline{8}, 2$	$0, \underline{6}$	$\underline{10}, 3$

Best-reply payoffs are underlined for convenience.

(c) The payoff table on the left applies. Recall R is strictly dominated. Two pure Nash, (T, L) and (B, C). By genericity there must be a fully mixed one too. Suppose player 1 plays $pT+(1-p)B$. Then player 2 is indifferent between L and C iff $5p+2(1-p) = p+6(1-p)$, i.e., $p = \frac{1}{2}$. Analogously, if player 2 mixes $qL+(1-q)C$ then player 1 is indifferent between T and B iff $5q + 10(1 - q) = 8q$ i.e., $q = 10/13$. The mixed Nash equilibrium is $\left(\frac{1}{2}T + \frac{1}{2}B, \frac{10}{13}L + \frac{3}{13}C\right)$ with payoffs $(\frac{50}{13} \approx 3.846, 3.5)$.

(d) The payoff table on the right applies. Recall R is dominated. No pure Nash, only a fully mixed one. Player 1 mixes as in part (c) because player 2's payoffs are unchanged. It is easy to check that player 2 mixes the same way as in (c) as well. Hence the unique mixed Nash equilibrium is $\left(\frac{1}{2}T + \frac{1}{2}B, \frac{10}{13}L + \frac{3}{13}C\right)$. The payoffs are $(\frac{80}{13} \approx 6.154, 3.5)$.

(e) Again, the payoff table on the right applies. Yes, this can be done using the subgame-perfect threat of reverting to stage-game mixed Nash forever upon deviation. After Heads player 1 is tempted to deviate to B to gain $8-5 = 3$, but if she does so she will lose on average $\frac{1}{2}10+\frac{1}{2}5-\frac{80}{13} \approx 1.346$ per period forever. After Tails player 2 is tempted to deviate to C to gain $6 - 3 = 3$, but then he loses $\frac{1}{2}5 + \frac{1}{2}3 - 3.5 = 0.5$ per period forever. Neither is worth it for δ close to 1.